Width vs Depth Under a Fixed Parameter Budget: An Empirical Study of the UAT in Practice

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1 Introduction

The Universal Approximation Theorem (UAT) guarantees that a sufficiently wide single-hidden-layer network can approximate any continuous function to arbitrary precision. What it does not say is whether gradient descent can *find* those weights under a realistic training budget.

This project started from a simple observation: when approximating $x\sin(x)+\cos(x^2)$ with a single hidden layer of 166 neurons, the network converged to MSE ≈ 0.487 and stayed there — regardless of how long training ran or which optimizer was used. Meanwhile, a 3-layer network with 13 neurons per layer (roughly the same number of parameters) approximated the same function to MSE $< 10^{-4}$.

This paper systematically studies that gap. We built a neural network framework from scratch using only NumPy — no PyTorch, no TensorFlow — and ran four ablation experiments across architectures, optimizers, function complexities, training budgets, and random seeds. The central finding is that the failure of wide-shallow networks on certain oscillatory functions is not a seed artifact or an optimizer choice — it is a deterministic, geometry-level property of the loss landscape under a single hidden layer.

2 Background

Definition 1 (Single-Hidden-Layer Network)**.** *A network with n hidden neurons computes*

$$F(\mathbf{x}) = \sum_{j=1}^{n} \alpha_j \tanh(w_j x + b_j)\,, \tag{1}$$

where $w_j, b_j, \alpha_j \in \mathbb{R}$ are learned parameters.

Theorem 1 (Cybenko 1989, Hornik 1991)**.** *For any continuous f on a compact set $K \subset \mathbb{R}^d$ and any $\varepsilon > 0$, there exists a single-hidden-layer network F with a continuous sigmoidal activation such that $\sup_{\mathbf{x}\in K} |f(\mathbf{x}) - F(\mathbf{x})| < \varepsilon$.*

Remark 1. *The UAT is an existence result. It guarantees that weights achieving ε-approximation exist — it says nothing about whether gradient descent finds them under a fixed training budget. This paper studies that gap empirically.*

3 Implementation

The entire framework is implemented from scratch in NumPy. No automatic differentiation library is used at any point.

3.1 Autograd Engine

A `Tensor` class wraps NumPy arrays and builds a computation graph dynamically. Each operation registers a `_backward` closure. Calling `.backward()` on the loss triggers a topological sort and propagates gradients in reverse:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}} \mathrel{+}= \texttt{\_backward}\left(\frac{\partial \mathcal{L}}{\partial \,\mathrm{out}}\right). \tag{2}$$

A key implementation detail: when bias $\mathbf{b}$ of shape $(m,)$ is broadcast to (N, m) in the forward pass, the gradient must be summed back over the batch dimension:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}} = \sum_{i=1}^{N} \frac{\partial \mathcal{L}}{\partial z_i}. \tag{3}$$

Without this "unbroadcast" step, gradient shapes mismatch and accumulation is silently incorrect.

3.2 Gradient Verification

Correctness was verified by two independent methods before any experiment was run:

Numerical gradient check (central difference, $h = 10^{-5}$): all weight gradients matched to relative error $< 10^{-10}$, bias gradients to $< 10^{-3}$ (floating-point noise at this scale).

PyTorch comparison: identical weights and inputs passed through both engines. All gradients agreed to $< 3 \times 10^{-8}$ absolute difference.

3.3 Components

Table 1: Framework components implemented from scratch.

Component	Description
`engine/tensor.py`	Tensor class, autograd, topological sort
`engine/ops.py`	NumPy forward/backward math primitives
`layers/dense.py`	Fully connected layer, Xavier init
`layers/activations.py`	ReLU, Sigmoid, Tanh
`layers/loss.py`	MSE loss
`network/model.py`	Sequential container
`network/optimizer.py`	SGD, SGD+Momentum, Adam

3.4 Architectures

Four architectures are studied throughout. All have approximately equal parameter counts (≈ 500), making comparisons fair:

Table 2: Architectures and parameter counts used in all experiments.

Architecture	Shape	Params
Wide-Shallow	$1 \to 166 \to 1$	499
Balanced	$1 \to 24 \to 24 \to 1$	501
Deep-Narrow	$1 \to 13 \to 13 \to 13 \to 1$	495
Very-Deep	$1 \to 10 \to 10 \to 10 \to 10 \to 1$	499

3.5 Target Functions

All experiments use $x \in [-\pi, \pi]$ sampled at 200 uniformly spaced points. Four target functions are used, ordered by oscillatory complexity:

$$f_1(x) = \sin(x), \tag{4}$$

$$f_2(x) = x \sin(x), \tag{5}$$

$$f_3(x) = x \sin(x) + \cos(x^2), \tag{6}$$

$$f_4(x) = \sin(3x)\, e^{-0.1x^2} + 0.1x. \tag{7}$$

f_3 has two overlapping oscillatory components with different frequencies — it is the primary stress-test function throughout.

4 Experiments and Results

4.1 Experiment A: Width vs Depth at Equal Parameter Budget

Setup. All four architectures trained with Adam ($\eta = 0.01$, 3000 epochs) on all four target functions. Parameter counts equalized to ≈ 500.

Results.

Table 3: Final MSE by architecture and function (3000 epochs, Adam).

Architecture	f_1	f_2	f_3	f_4
Wide-Shallow	5.81×10^{-5}	3.56×10^{-4}	4.87×10^{-1}	2.84×10^{-1}
Balanced	1.22×10^{-5}	1.81×10^{-5}	1.46×10^{-4}	4.97×10^{-5}
Deep-Narrow	3.55×10^{-6}	4.88×10^{-6}	1.41×10^{-4}	4.54×10^{-5}
Very-Deep	5.09×10^{-6}	4.80×10^{-5}	3.66×10^{-3}	2.12×10^{-5}

What this means. Wide-Shallow fails catastrophically on f_3 and f_4 — MSE ≈ 0.487 on f_3, which is the same as predicting the mean of the target function. Deep-Narrow achieves 1.41×10^{-4} on the same function with 3000x fewer effective parameters per layer. The difference is not parameter count — it is architecture shape.

Notably, Wide-Shallow's loss curve on f_3 goes flat within 50 epochs and never moves: the network finds a smooth, lazy local minimum (predicting the mean trend) and Adam cannot escape it because the gradient signal from the residual oscillation is too weak relative to the already-saturated tanh units.

4.2 Experiment B: Optimizer Comparison

Setup. All four architectures trained with SGD, SGD+Momentum, and Adam on f_3 (the hardest function). 3000 epochs each.

Results.

Table 4: Final MSE by architecture and optimizer on $f_3 = x\sin(x) + \cos(x^2)$.

Architecture	SGD	SGD+Momentum	Adam
Wide-Shallow	4.91×10^{-1}	2.97×10^{-1}	4.87×10^{-1}
Balanced	6.90×10^{-2}	3.01×10^{-4}	1.19×10^{-4}
Deep-Narrow	1.77×10^{-1}	6.82×10^{-4}	8.48×10^{-5}
Very-Deep	1.38×10^{-1}	1.12×10^{-2}	2.75×10^{-4}

What this means. Wide-Shallow stays stuck (MSE > 0.29) under all three optimizers. This rules out the hypothesis that the failure is an Adam artifact. The plateau is architectural, not optimizer-specific.

A secondary finding: plain SGD fails on every architecture (best result: 6.9×10^{-2}). Momentum — either via SGD+Momentum or Adam — is necessary for meaningful convergence on these functions. The loss curves show a characteristic staircase pattern under momentum-based methods: repeated spike-and-drop events corresponding to plateau escapes. Plain SGD shows smooth but permanently high loss — it never escapes.

4.3 Experiment C: Minimum Width Required per Function

Setup. Single-hidden-layer networks of widths $\{2, 4, 8, 16, 32, 64, 128, 256, 512\}$ trained on each function (3000 epochs, Adam). Threshold: MSE $< 10^{-4}$.

Results.

Table 5: Minimum width for single-hidden-layer network to reach MSE $< 10^{-4}$.

Function	Min width	Notes
$\sin(x)$	4	Reliable
$x\sin(x)$	32	Only width=32 passes; 64-512 all fail
$x\sin(x) + \cos(x^2)$	—	Never reached at any width
$\sin(3x)e^{-0.1x^2}$	32	Fails again at width ≥ 64

What this means. $f_3 = x\sin(x) + \cos(x^2)$ never crosses the threshold at any width from 2 to 512. The MSE is approximately constant at ≈ 0.489 across all widths — increasing width does not help at all. This confirms the failure is not a capacity problem (512 neurons is far beyond what the UAT would require). It is a loss landscape problem: every initialization finds the same bad local minimum.

The non-monotonic behavior on f_2 and f_4 is also notable: width=32 passes but width=64, 128, 256, 512 all fail. Performance degrades monotonically for width > 32, suggesting that larger single-layer networks have increasingly difficult optimization landscapes for oscillatory targets.

4.4 Experiment D: Training Budget Sensitivity

Setup. Wide-Shallow and Deep-Narrow trained for 15,000 epochs (5× the standard budget) on $\sin(x)$ and f_3.

Results.

Table 6: Results at 15,000 epochs vs 3,000 epochs.

Architecture	Function	Final MSE (15k)	First PASS
Wide-Shallow	$\sin(x)$	1.67×10^{-5}	epoch 1180
Wide-Shallow	f_3	9.02×10^{-4}	Never
Deep-Narrow	$\sin(x)$	4.04×10^{-5}	epoch 229
Deep-Narrow	f_3	1.41×10^{-4}	epoch 1998

What this means. Wide-Shallow sat at MSE ≈ 0.487 for 11,000 consecutive epochs before finally escaping the plateau at epoch $\approx 11,000$ — only to land in a different bad region (MSE 9×10^{-4}), still far above threshold. Five times the training budget is not enough to rescue a single-layer network on this function. The plateau is not a patience problem — it is a structural entrapment.

Deep-Narrow crosses the threshold at epoch 1998 but oscillates around it for the remainder of training, suggesting the function is at the edge of what this architecture can stably represent within this parameter budget.

4.5 Experiment E: Seed Robustness

Setup. All four architectures trained across 5 random seeds $\{0, 7, 21, 42, 99\}$ on $\sin(x)$ and f_3. 3000 epochs, Adam.

Results.

Table 7: Pass rate and mean ± std MSE over 5 seeds.

Function	Architecture	Pass rate	Mean MSE	Std
$\sin(x)$	Wide-Shallow	3/5	2.84×10^{-4}	4.09×10^{-4}
	Balanced	5/5	1.37×10^{-5}	1.05×10^{-5}
	Deep-Narrow	3/5	1.78×10^{-4}	2.45×10^{-4}
	Very-Deep	5/5	5.94×10^{-6}	3.92×10^{-6}
f_3	Wide-Shallow	0/5	4.88×10^{-1}	8.21×10^{-4}
	Balanced	2/5	1.99×10^{-4}	1.46×10^{-4}
	Deep-Narrow	3/5	1.28×10^{-4}	8.36×10^{-5}
	Very-Deep	0/5	1.56×10^{-3}	1.47×10^{-3}

What this means. Three observations stand out.

First, Wide-Shallow's failure on f_3 is deterministic: 0/5 seeds, mean MSE 4.88×10^{-1}, std 8.21×10^{-4}. The near-zero standard deviation means every random initialization converges to essentially the same bad local minimum. This is not stochastic bad luck — it is a fixed point of the loss landscape under a single hidden layer.

Second, Deep-Narrow is the only architecture that reliably attempts f_3, with 3/5 seeds passing and the lowest mean MSE among all architectures on this function. It is also the only architecture with consistent performance across both easy and hard functions.

Third, Very-Deep (4 layers) performs worse than Deep-Narrow (3 layers) on f_3 — 0/5 vs 3/5, with much higher variance. This suggests that for this parameter budget, 4 narrow layers introduces vanishing gradient issues that offset the benefit of additional compositional depth. Three layers appears to be the sweet spot.

5 Discussion

Taken together, the five experiments support a specific, falsifiable claim about the gap between UAT theory and practical learnability:

> *Under a fixed parameter budget and training schedule, single-hidden-layer networks exhibit a deterministic failure mode on dual-frequency oscillatory targets — converging to the same local minimum (MSE ≈ 0.487) across all random seeds and all optimizers tested. This failure is not a capacity limitation (it persists at widths up to 512) and not a training-time limitation (it persists up to 15,000 epochs). It is a property of the loss landscape geometry under a single hidden layer: the flat local minimum corresponding to the mean trend of the function is a fixed attractor that gradient descent cannot escape.*

Depth breaks this attractor. A 3-layer network with 13 neurons per layer — same parameter count, radically different shape — escapes the plateau on 3/5 seeds and achieves mean MSE 1.28×10^{-4}. The mechanism is compositional: each layer can specialize in a sub-feature (local phase, amplitude envelope, frequency component), and the staircase loss behavior observed in training curves suggests repeated plateau escapes driven by momentum accumulation in earlier layers propagating reorganization signals to later ones.

The non-monotonic behavior in Experiment C — where width=32 passes but width=64, 128, 256, 512 all fail on f_2 — suggests that the optimization landscape becomes increasingly difficult with width, not easier. Larger single-layer networks have more parameters that can saturate in concert, creating broader and deeper flat regions in the loss surface.

Limitations. All experiments use a single domain ($x \in [-\pi, \pi]$), a single activation (tanh), and a fixed optimizer (Adam with $\eta = 0.01$) except where optimizer is the independent variable. The findings may not generalize to other activations (e.g., ReLU, GELU), other domains, or other learning rate schedules. The Hessian eigenvalue analysis suggested by the loss landscape results is left for future work.

6 Conclusion

We presented an empirical study of the gap between the UAT's existence guarantee and practical learnability under a fixed compute budget. Five experiments across architectures, optimizers, function complexities, training budgets, and random seeds consistently support the same conclusion: width alone does not bridge this gap for oscillatory functions with multiple frequency components. Depth — specifically 3 layers with narrow width — is both necessary and sufficient to escape the deterministic failure mode that

traps single-hidden-layer networks, within the parameter and compute budget studied here.

The entire framework — autograd engine, layers, optimizers, and all experiments — is implemented from scratch in NumPy and is available at https://github.com/Sage-Vortex/Neural-Network-from-scratch-Universal-Approximation-Theorem.

References

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